VLC MEDIA PLAYER

The Problem VLC Was Created to Solve

During initial time, there were limited tools to play media, and those were not easy to use because they only supported few formats, people had to download additional codecs which were unsafe and hard to download. This all work to just play a media was very frustrating.

At Ecole Centrale Paris there was a group of students, whom also faced these problems with additional one.

They wanted a network running through there campus, to share and run content, so all the students could access the content and run that media. But obviously this was not possible using those tools, because they were not flexible enough, also it was expensive and student couldn't afford it.
To solve this problem students built their own system which was later known as VLC media player.
Their goal was to create a tool that could handle multiple media formats without requiring additional downloads and could also support streaming over a network

Instead of keeping it private, they released it as open-source software so that others could use, modify, and improve it.

This decision was important because it allowed VLC to grow with contributions from developers around the world. Over time, it became a universal media player capable of playing almost any file format. The original frustration of limited compatibility and restricted software access directly shaped VLC's design philosophy — simplicity, flexibility, and openness.

The License — What It Actually Says

VLC Media Player is released under the GNU General Public License (GPL). This is one of the most widely used open-source licenses, designed to ensure that software remains free and open for all users.

The GPL is a copyleft license, meaning that any modified version of the software must also remain open-source when distributed. This ensures that the freedom of the software is preserved over time.

1. The Four Freedoms of Free Software

The Four Freedoms:

1. Freedom to run the program for any purpose
2. Freedom to study how the program works (access to source code)
3. Freedom to modify the program
4. Freedom to distribute copies, including modified versions

Does VLC provide all four freedoms?

VLC fully supports all four freedoms.
Users can:

- Use VLC freely
- Access its source code
- Modify it
- Share it with others

<u>Can a Company Modify and Sell VLC Without Sharing Changes?</u>

No they cannot.

According to GPL, they can sell and modify VLC, but they must share the source code, and has to release their modifications under the same GPL license.

Difference between GPL and MIT

GPL priotizes openness or they force to openness while MIT allows privatization, it is more flexible.

I prefer MIT over GPL, because I will feel less restrictions and questionable

“FREE AS A BEER OR FREE SPEECH”

I think it is FREE as in FREE SPEECH, because we can modify it, use it freely, while we are not harmed because of its excessive use as FREE IN BEER.

The Ethics of Open Source

Open source software is not only about making code available — it represents a philosophy about how knowledge should be shared in society. Projects like VLC Media Player show how collaboration and openness can create powerful tools used by millions of people worldwide.

However, open source also raises important ethical questions about fairness, ownership, and profit.

1. Should All Software Be Open Source?

Open source promotes:

- Transparency
- Collaboration
- Innovation

When source code is open:

- Anyone can inspect it for security issues
- Developers can improve it
- Knowledge becomes accessible to everyone

For example, VLC allows users to play almost any media file without restrictions. If it were closed-source, users would depend on a company's decisions and limitations.

While Not all software should be open source

- Companies invest money and need profit
- Some systems require privacy (e.g., banking, defense)
- Open code can sometimes be misused

2. Is It Ethical for Companies to Profit from Open Source?

Many companies like Red Hat, Google, and Meta build products using open-source software like Linux.

It IS ethical

- Companies add value (support, services, improvements)
- They often contribute back to the community
- They help scale open-source projects globally

Example:

- Red Hat sells support for Linux, not the software itself

👉 This creates a sustainable ecosystem.

It can be UNETHICAL

- Companies may profit from community work without fair contribution
- Developers who built the original software may not get rewarded
- Power shifts from community to corporations

3. “Standing on the Shoulders of Giants”

Building new ideas based on previous knowledge.

In software:

- Developers don’t start from zero
- They use existing libraries, tools, and frameworks

For example:

- VLC uses codecs and libraries created by others
- It builds on years of previous work

Does Open Source Help or Hurt Innovation?

It ENABLES innovation

- Faster development
- Less duplication of work
- More collaboration

Developers can focus on improving ideas instead of reinventing basics.

Linux Footprint of VLC Media Player

1. Installation Method

VLC can be installed using Linux package managers, making it very easy to deploy.

Using APT (Debian/Ubuntu systems):

sudo apt update
sudo apt install vlc

Other methods:

- Snap package: sudo snap install vlc
- Flatpak package
- No need to manually compile from source
- Reduces errors and saves time

2. Key Directories

After installation, VLC files are distributed across standard Linux directories:

Important directories:

- /usr/bin/vlc → Main executable file
- /usr/lib/vlc/ → Libraries and plugins
- /usr/share/vlc/ → Icons, UI files, resources

- ~/.config/vlc/ → User-specific configuration

What is what

- /usr/bin/ → contains executable programs
- /usr/lib/ → contains libraries needed by VLC
- /usr/share/ → contains shared data like themes
- ~/.config/ → stores personal settings

3. User and Group

VLC runs as:

The **current logged-in user**
Not as root (administrator)

Why is this important?

Running as a normal user:

- Prevents system-wide damage
- Limits access to sensitive files
- Improves security

Example:

If VLC crashes or is exploited:
It cannot harm the entire system because it doesn't have root access.

This is a **basic Linux security principle**.

4. Service Management

VLC is **not a background system service** like a web server.

It is a **user-level application**, meaning:

- It starts when the user runs it
- It stops when the user closes it

Commands:

vlc

cvlc

- vlc → GUI version
- cvlc → command-line version

Can it run as a service?

✓ Yes (advanced use):

- VLC can be used for streaming as a background service

Example:

cvlc video.mp4 --loop

5. Update Model

VLC updates are managed through the Linux package system.

Commands:

sudo apt update
sudo apt upgrade

How updates work:

1. Developers update VLC source code
2. Linux distribution maintainers package it
3. Updates are pushed to repositories
4. Users receive updates via package manager

Why this is important:

- Security vulnerabilities are fixed quickly
- Users don’t need to manually check for updates
- System remains stable and secure

6. Security Perspective (Add this for extra marks)

- VLC does not run as root
- Uses system libraries (trusted dependencies)
- Updated through verified repositories

This reduces:

- Malware risk
- Unauthorized access

- System instability

The FOSS Ecosystem

VLC Media Player does not exist independently but is part of a larger open-source ecosystem where multiple tools and communities interact. It depends on several important open-source libraries such as FFmpeg for handling audio and video processing, along with codec libraries like x264 and x265 that enable efficient compression and playback of media files. Instead of building everything from scratch, VLC integrates these components, which reflects the collaborative nature of open-source development. At the same time, VLC has enabled various applications such as media streaming systems, automated playback tools, and educational use cases, especially through its command-line interface. Although it is not directly part of the traditional LAMP stack, it plays an important supporting role in web-based media delivery by helping stream audio and video content. The project is supported by a global community of developers and contributors who collaborate through platforms like Git repositories, forums, and mailing lists, and it is governed by the VideoLAN, which ensures that

development remains open, organized, and aligned with the principles of free and open-source software.

Open Source vs Proprietary (VLC Comparison)

proprietary alternative: Windows Media Player

Dimension	**Open Source (VLC Media Player)**	**Proprietary (Windows Media Player)**
Cost	Completely free, no ads or paid features	Free but limited, some features tied to ecosystem
Security	Source code is open, anyone can audit it	Source code is closed, only company can audit
Support & Reliability	Community-based support, forums and updates	Official support from Microsoft, more structured
Freedom to Modify	Full freedom to modify and redistribute	No modification allowed

Dimension	Open Source (VLC Media Player)	Proprietary (Windows Media Player)
Community vs Control	Community-driven (VideoLAN)	Corporate-controlled (Microsoft)

Conclusion

From this analysis, it is clear that open-source software like VLC Media Player provides significant advantages in terms of flexibility, transparency, and user freedom. Its ability to support a wide range of media formats without restrictions makes it highly practical for everyday use. However, proprietary software such as Windows Media Player offers more structured support and integration within a larger ecosystem, which can be beneficial for less technical users. Therefore, open source is not always superior in every aspect, but it provides greater control and independence to users.

In my opinion, VLC Media Player is highly suitable for real-world deployment because of its reliability, performance, and strong community support. It is especially useful in environments where flexibility and

cost-effectiveness are important. I would recommend using VLC over proprietary alternatives in most cases. Additionally, I would be interested in contributing to open-source projects like VLC in the future, as it promotes collaboration, learning, and continuous improvement of software.